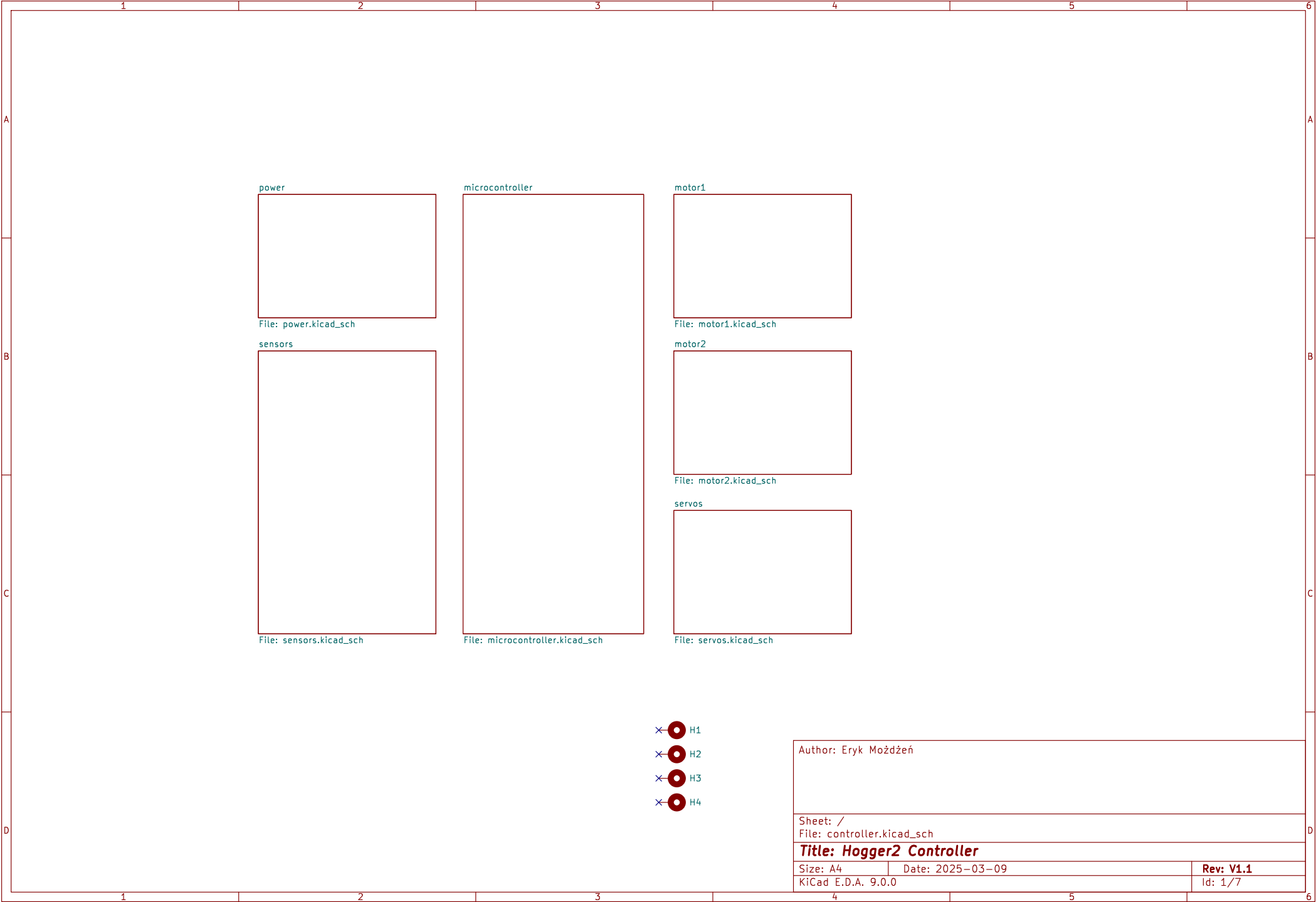
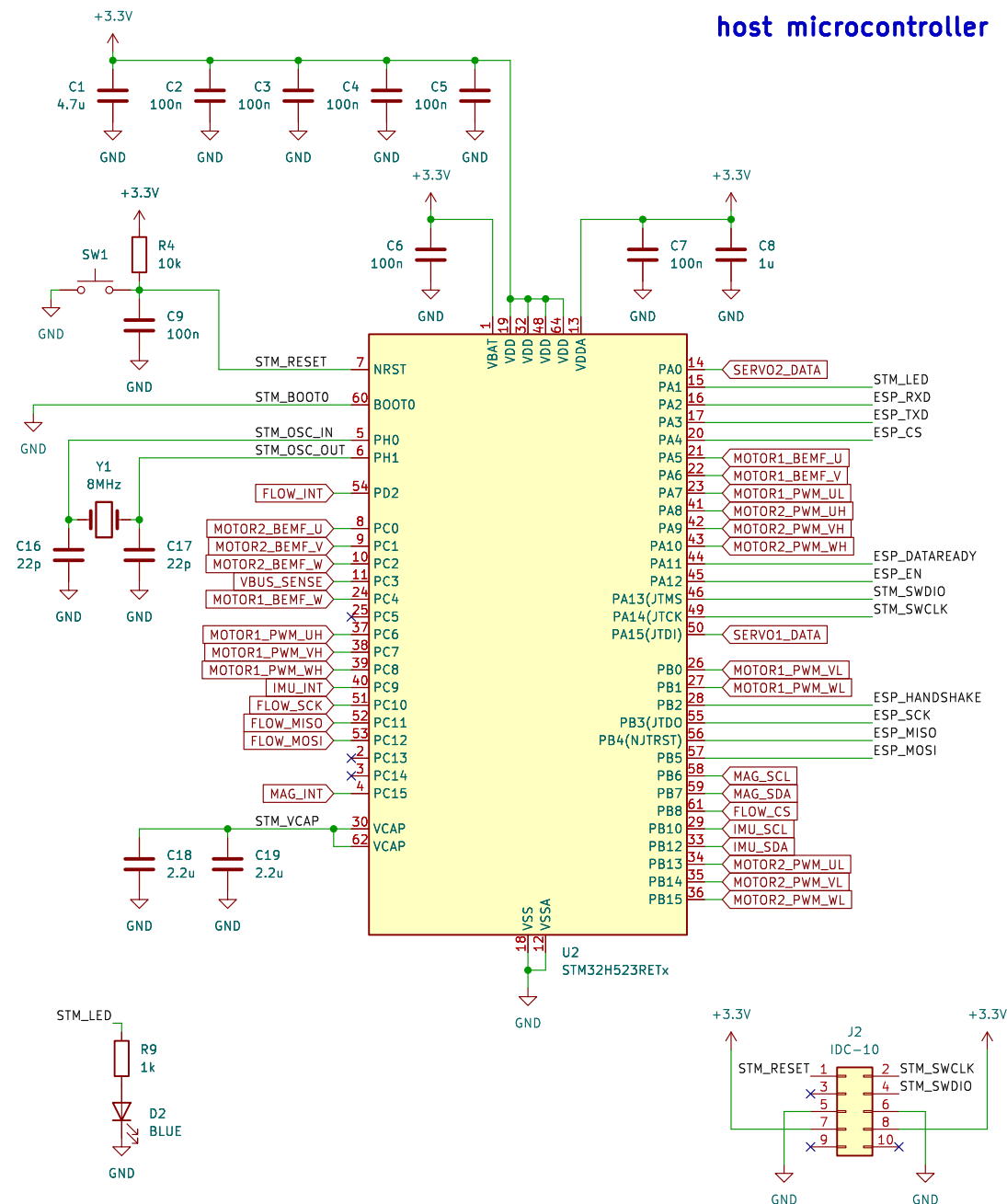
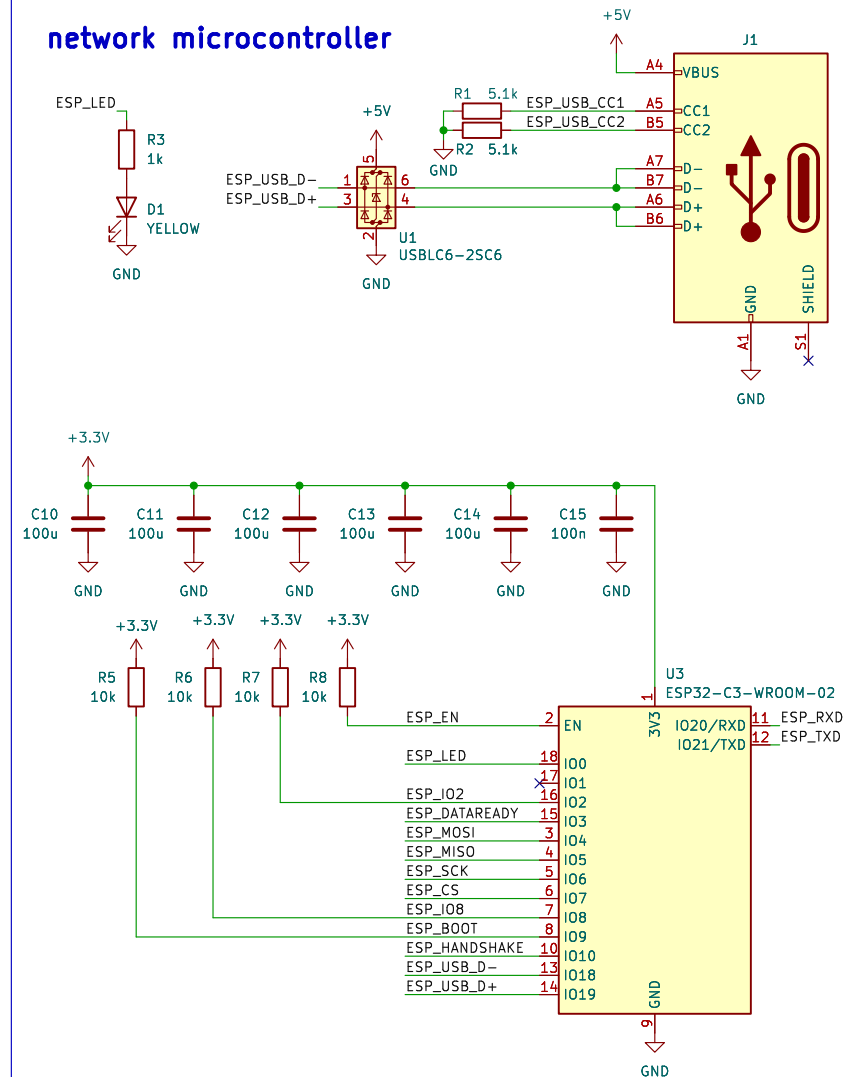




host microcontroller



network microcontroller



Author: Eryk Mozdzeń

Sheet: /microcontroller/
File: microcontroller.kicad_sch

Title: Hogger2 Controller

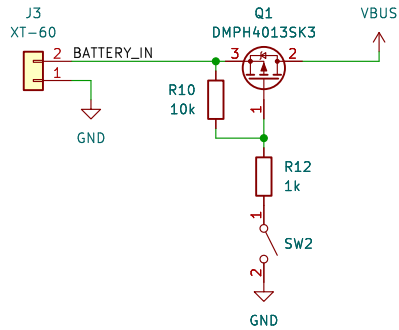
Size: A4 Date: 2025-04-12

KiCad E.D.A. 9.0.0

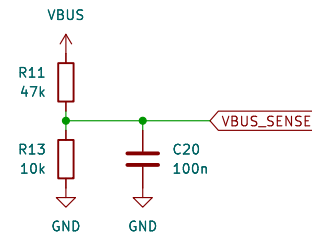
Rev: V1.1

Id: 2/7

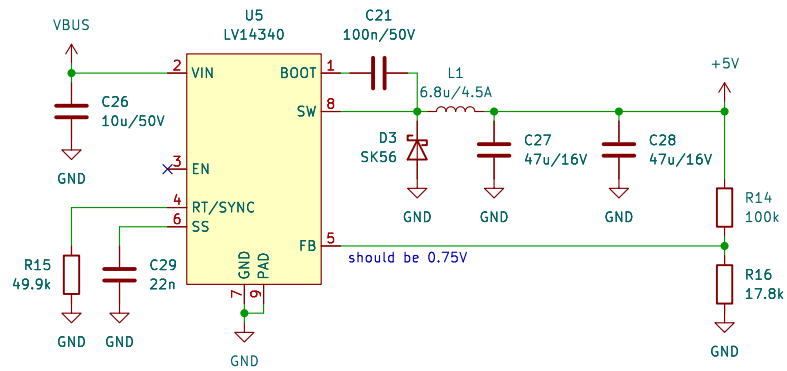
power input



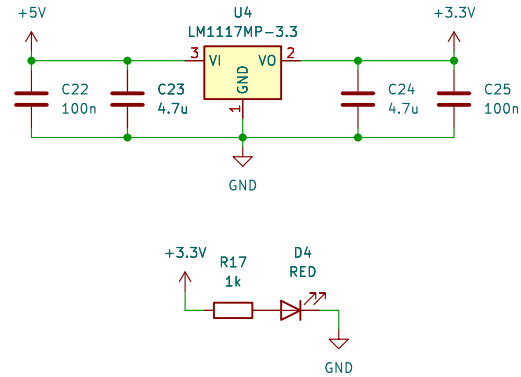
VBUS voltage measurement



5V domain



3.3V domain



Author: Eryk Możdżeń

Sheet: /power/
File: power.kicad_sch

Title: Hogger2 Controller

Size: A4 Date: 2025-03-09

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Rev: V1.1

Id: 3/7

accelerometer & gyroscope

accelerometer & gyroscope

U6 MPU-6050

Power supply: +3.3V

Capacitors: C30 100n, C31 100n, C34 100n, C35 2.2n

Resistors: R18 4.7k, R19 4.7k

I2C address: 0x68 (7-bit)

IMU_SDA, IMU_SCL, IMU_INT, IMU_CPOUT, IMU_REGOUT

FSYNC, CLKIN, VLOGIC, VDD, GND

optical flow

+3.3V

C32 10u/50V

C33 100n

GND

GND

J4

1 2 3 4 5 6 7

FLOW_CS

FLOW_SCK

FLOW_MOSI

FLOW_MISO

FLOW_INT

GND

connected to external PMW3901 dev-board

magnetometer

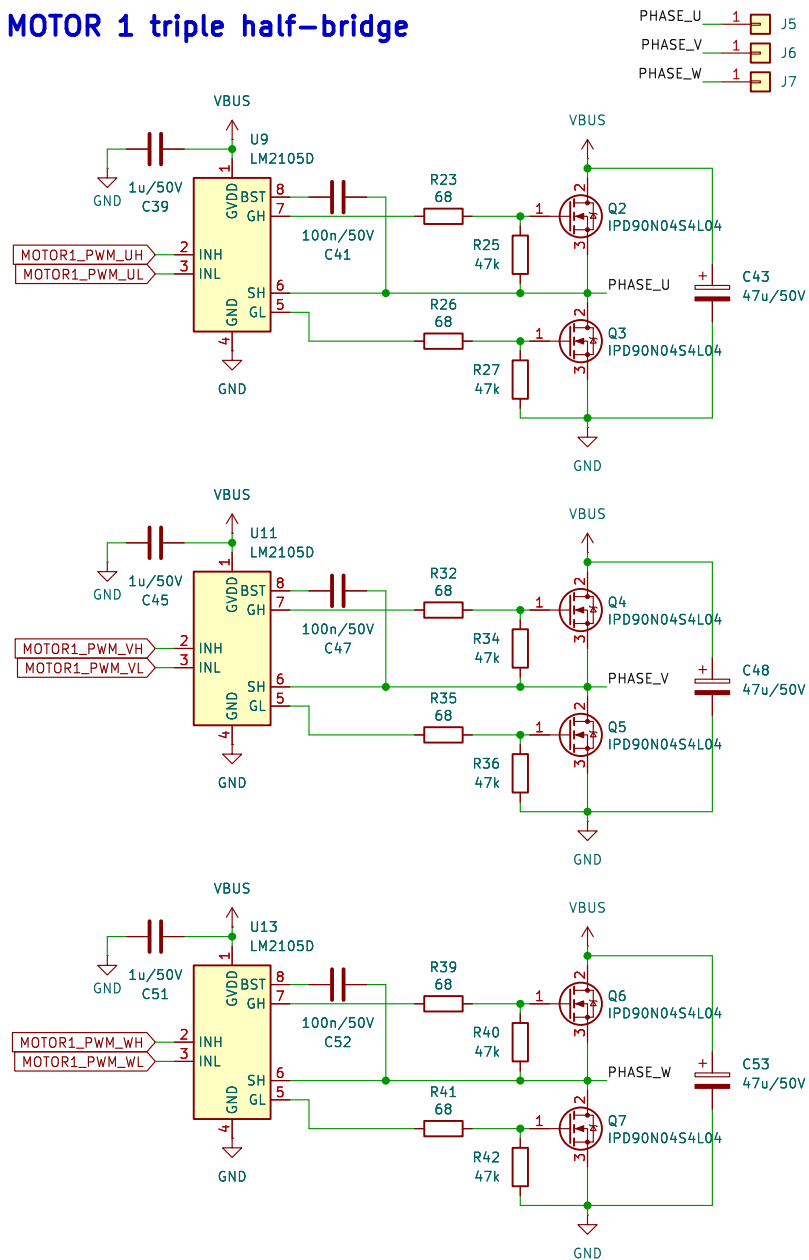
The diagram shows the HMC5883L magnetometer module (U7) connected to a +3.3V supply and ground. The module's pins are connected as follows:

- Power and Ground:**
 - Pin 13 (VDDIO) is connected to +3.3V.
 - Pin 4 (S1) is connected to +3.3V.
 - Pin 2 (VDD) is connected to +3.3V.
 - Pin 9 (GND) is connected to ground.
 - Pin 11 (GND) is connected to ground.
- I2C Interface:**
 - Pin 16 (SDA) is connected to +3.3V.
 - Pin 1 (SCL) is connected to +3.3V.
 - Pin 15 (DRDY) is connected to +3.3V.
- Configuration Pins:**
 - Pin 8 (MAG_SETP) is connected to +3.3V.
 - Pin 12 (MAG_SETC) is connected to +3.3V.
 - Pin 10 (MAG_C1) is connected to +3.3V.

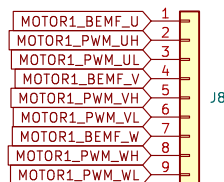
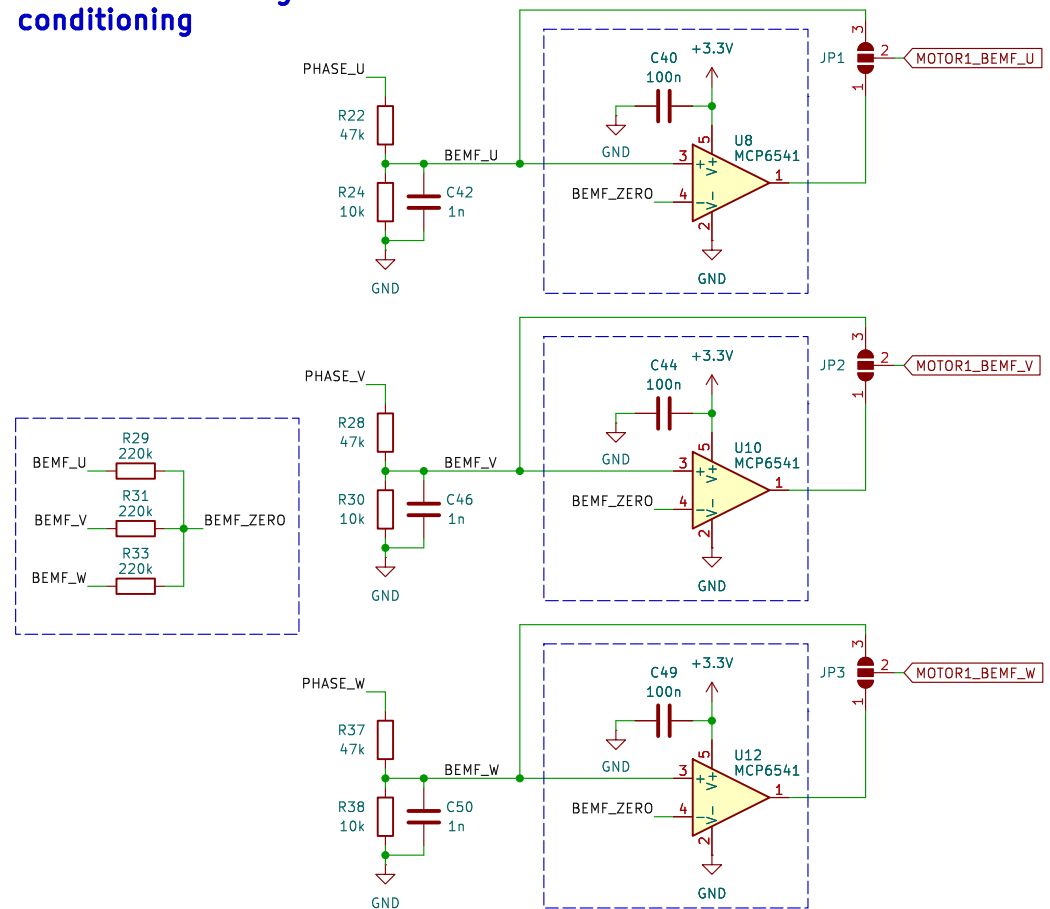
External components include capacitors C37 (220nF), C38 (4.7uF), C36 (100nF), and resistors R20 (4.7k), R21 (4.7k). The I2C address is 0x1E (7-bit).

Id: 4/7

MOTOR 1 triple half-bridge



MOTOR 1 BEMF signals conditioning



Author: Eryk Mozdzeń

Sheet: /motor1/
File: motor1.kicad_sch

Title: Hogger2 Controller

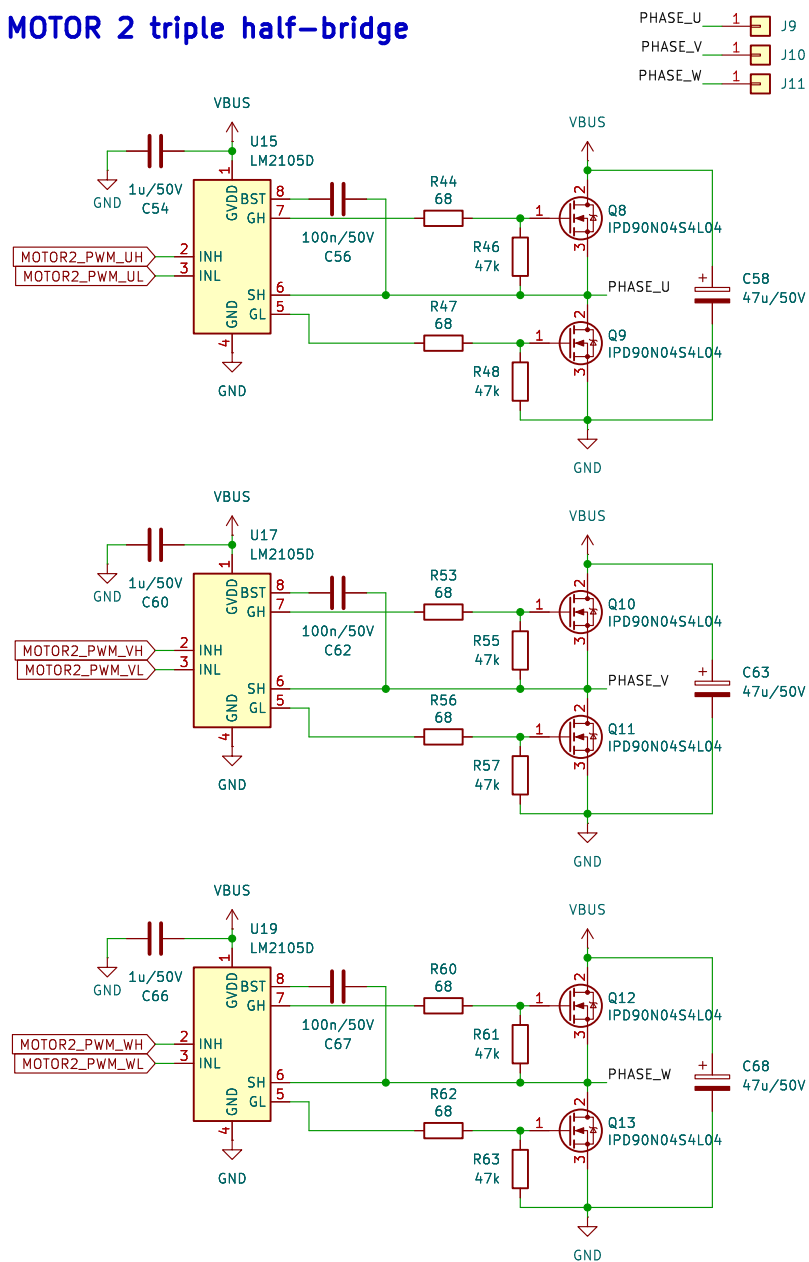
Size: A4 Date: 2025-03-09

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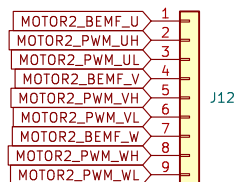
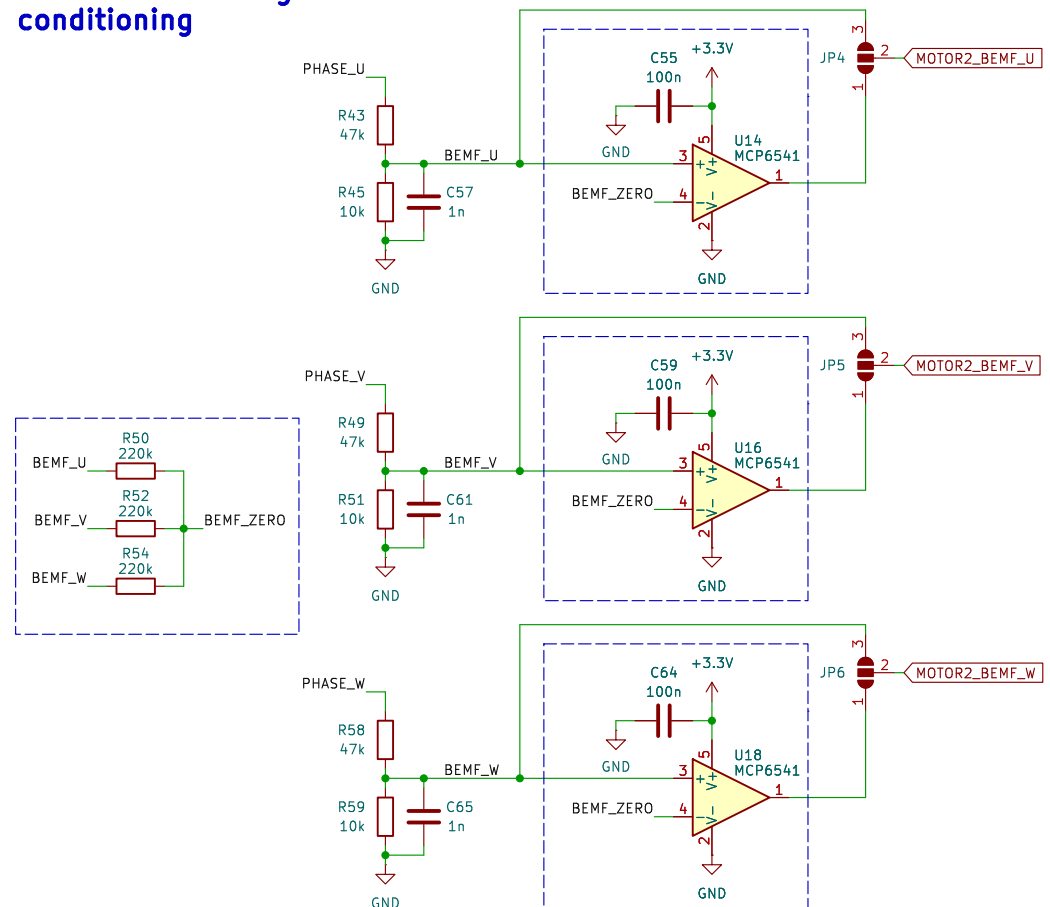
Rev: V1.1

Id: 7/7

MOTOR 2 triple half-bridge



MOTOR 2 BEMF signals conditioning



Author: Eryk Możdżeń

Sheet: /motor2/
File: motor2.kicad_sch

Title: Hogger2 Controller

Size: A4 Date: 2025-03-09

KiCad E.D.A. 9.0.0

Rev: V1.1

Id: 7/7

